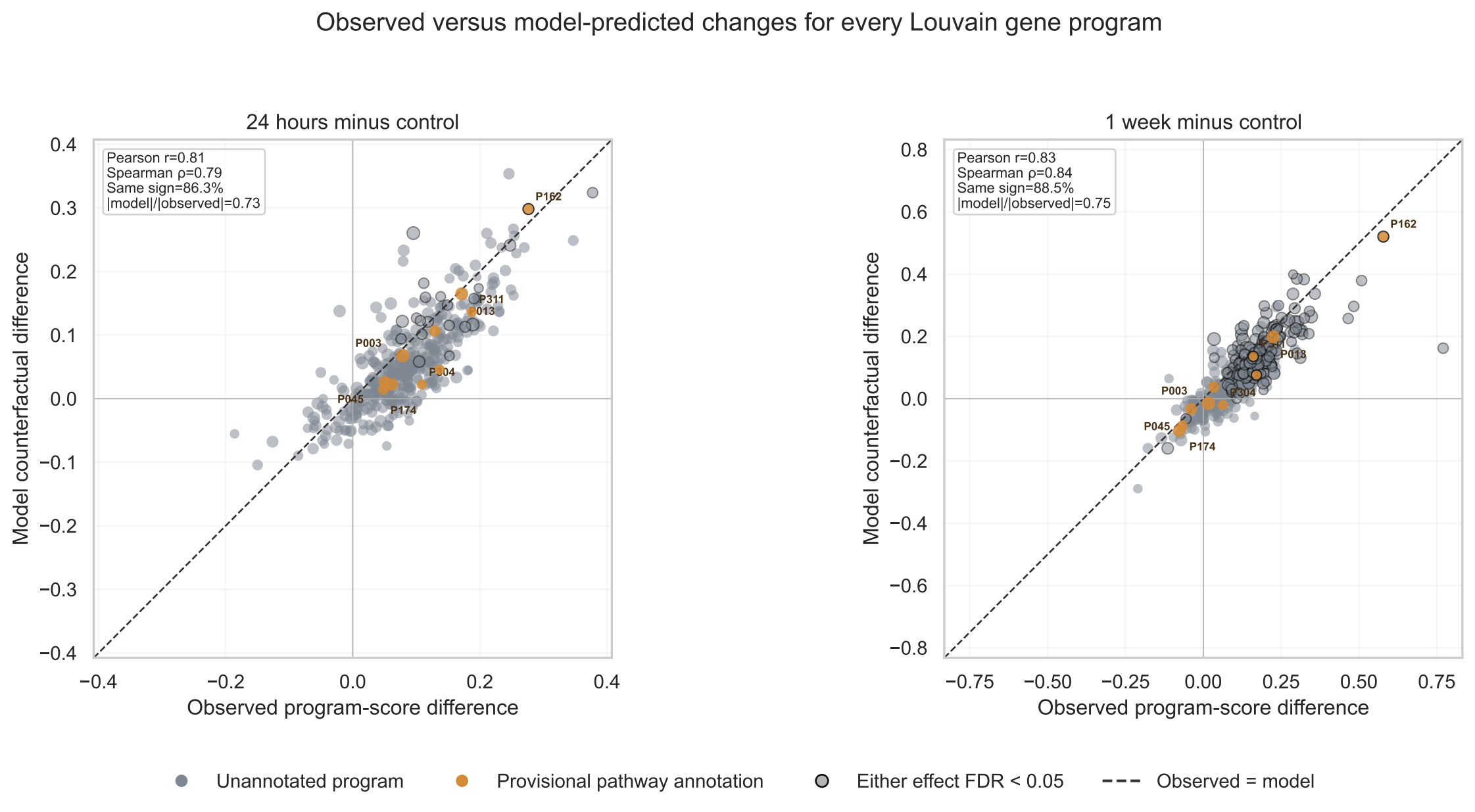

Observed versus model-predicted changes for every Louvain gene program
24 hours minus control
Pearson r=0.81
Spearman ρ=0.79
Same sign=86.3%
|model|/|observed|=0.73
Model counterfactual difference
Observed program-score difference
P162
P311
P013
P003
P304
P045
P174
1 week minus control
Pearson r=0.83
Spearman ρ=0.84
Same sign=88.5%
|model|/|observed|=0.75
Model counterfactual difference
Observed program-score difference
P162
P018
P003
P304
P045
P174
Unannotated program
Provisional pathway annotation
Either effect FDR < 0.05
Observed = model